

Discovery Executive Summary

Project: test-discovery-001 · Generated: 12/08/2026, 20:17:23

EXECUTIVE SUMMARY

This report consolidates the overall ratings, key findings, and recommended actions from the 1 discovery analysis run across this codebase (frontend and backend). Each section below reproduces that analysis's executive view; full evidence and diagrams live in the individual reports.

Portfolio Overview

#	Analysis	Overall Rating
1	Architecture & Design Analysis	—

1. Architecture & Design Analysis

EXECUTIVE SUMMARY

The Klearcom Voice Observability Platform is a three-layer monorepo: a Laravel 12 PHP backend, a React 19 TypeScript SPA, and a Node.js Express dev-API mock. Overall architectural health is **Moderate**. The most severe hotspot is a pervasive **missing repository pattern**: every controller directly invokes Eloquent ORM methods — 32 direct ORM call-sites across 4 production controllers — with zero repository or data-access abstraction layer. A second critical issue is a **triply duplicated reachability formula** (identical `successRate` calculation present verbatim in `ConnectController`, `LegacyReportController`, `RealTimeTestService`, and `dev-api/realtime.js`), creating hidden change-amplification risk on any KPI definition change. The frontend has a **legacy class component** (`LegacyMonitorPoller.jsx`) with an intentional interval leak and no cleanup, mixed alongside modern functional components. No circular dependency cycles or god classes exceeding 1000 LOC were found. Addressing the repository pattern and duplicated domain logic is the highest-priority corrective action before the platform scales to additional carriers or regions.

§1.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	Good	Moderate	High Risk	Measured	Rating
H1	Fat Controllers	Avg LOC per controller	<150	150–300	>300	65 LOC avg (max 102)	Good
H2	Missing Service Layer	Controllers accessing repos/models directly	<10	10–20	>20	32 direct ORM call-sites across 4 controllers	Moderate
H3	Missing Repository Pattern	Direct DB access points outside repositories	<10	10–20	>20	32 Eloquent call-sites; 0 repositories exist	Moderate
H4	Circular Dependencies	Dependency cycles	0	1–3	>3	0 cycles detected	Good
H5	Shared Utility Abuse	Utility files with business logic	0	1–5	>5	1 (<code>LegacyDataMapper</code> using unsafe <code>extract()</code>)	Moderate
H6	Direct SQL in Controllers	ORM compliance %	>90%	60–90%	<60%	100% ORM (no raw SQL found)	Good
H7	God Classes	Classes/files >1000 LOC	0	1–3	>3	0 (largest: <code>dev-api/server.js</code> 276 LOC)	Good
H8	Domain Boundary Violations	Cross-domain access points	0	1–5	>5	4 (<code>LegacyReportController</code> + <code>RealTimeTestService</code> import both <code>Connect</code> and <code>Discovery</code> models)	Moderate
H9	Shared Database Coupling	Tables/collections shared across domains	<10%	10–30%	>30%	3 of 3 MongoDB collections shared between <code>Connect</code> and <code>Discovery</code> (100%)	Moderate
F1	Business Logic in Components	Avg LOC per component	<150	150–300	>300	81 LOC avg (max 222 <code>ConnectPage</code>)	Good
F2	Missing Frontend Service/Data Layer	Components with inline API calls	<10	10–20	>20	0 inline — all calls via <code>api/client.ts</code>	Good

F3	God / Oversized Components	Components >400 LOC	0	1–3	>3	0 (largest: <code>ConnectPage</code> 222 LOC)	Good
F4	Prop Drilling / Global State Abuse	Max prop-drilling depth	≤2	3–4	>4	≤2 levels	Good
F5	Legacy / Inconsistent Component Patterns	Legacy-pattern components	0	1–10	>10	2 (<code>LegacyMonitorPoller.jsx</code> class + interval leak; <code>LegacyDashboardWidget.tsx</code> no Error Boundary)	Moderate

§1.4 Actions Required

Hotspot	Action	Rating	Priority
H2 — Missing Service Layer	Create <code>ConnectAnalyticsService</code> (reachability formula) and <code>DashboardService</code> (KPI aggregation); eliminate 32 inline ORM calls from controllers and deduplicate the 4-copy reachability formula	Moderate	High
H3 — Missing Repository Pattern	Introduce <code>ConnectMonitorRepositoryInterface</code> , <code>ConnectCheckResultRepositoryInterface</code> , <code>DiscoveryJobRepositoryInterface</code> , <code>DiscoveryNodeRepositoryInterface</code> with Eloquent implementations; register in <code>AppServiceProvider</code> ; replace all static Eloquent calls	Moderate	High
H5 — Shared Utility Abuse	Replace <code>extract()</code> in <code>LegacyDataMapper</code> with explicit array key access; register class in Laravel DI container; rename to domain-specific presenter	Moderate	Medium
H8 — Domain Boundary Violations	Split <code>RealTimeTestService</code> into <code>ConnectTestService</code> and <code>DiscoveryTestService</code> ; split <code>LegacyReportController</code> into domain-scoped controllers; enforce single-domain imports via PHPStan	Moderate	Medium
H9 — Shared Database Coupling	Separate MongoDB collections per domain (<code>connect_transcripts</code> , <code>discovery_transcripts</code>); create domain-scoped <code>ConnectMongoAdapter</code> and <code>DiscoveryMongoAdapter</code>	Moderate	Medium
F5 — Legacy Component Patterns	Convert <code>LegacyMonitorPoller.jsx</code> to functional component with <code>useEffect</code> cleanup; wrap <code>LegacyDashboardWidget</code> in Error Boundary; add ESLint rule blocking class components	Moderate	Medium

§1.5 Expected Outcomes

- **Testable controllers and services:** With repository interfaces injected, all controllers can be unit-tested against mock repositories without a running database, reducing fixture overhead from database seeding to simple in-memory fakes.
- **Single source of truth for business rules:** Centralizing the reachability formula in `ConnectAnalyticsService` eliminates the current 4-copy divergence — future threshold changes require editing exactly one method in one file.
- **Independent domain evolution:** Splitting `RealTimeTestService` and `LegacyReportController` by bounded context means Connect and Discovery teams can refactor their persistence layers independently without coordinated cross-domain changes.
- **Safer MongoDB schema changes:** Domain-scoped collections allow TTL policies, index tuning, and schema evolution per domain without risk of a missing `module` filter silently returning another domain's records.
- **Consistent, leak-free frontend:** Migrating `LegacyMonitorPoller` to a functional component eliminates the interval leak; adding an Error Boundary to `LegacyDashboardWidget` prevents API unavailability from crashing the root app tree.

Reports saved to:

- `docs/discovery/01-architecture-design.md` — primary deliverable (PDF auto-generated by orchestration UI)
- `agent-runs/20260812T200500_sgnbbt/01-architecture-design.md` — pipeline artifact with run header{"stop_reason":"end_turn","session_id":"c72e5ab5-cef0-48ab-a4cb-201ee53a4d3e","total_cost_usd":1.8308634500000005,"usage":{"input_tokens":7,"cache_creation_input_tokens":27892,"cache_read_input_tokens":95114,"output_tokens":3996,"server_tool_use":{"web_search_requests":0,"web_fetch_requests":0},"service_tier":"standard","cache_creation":{"ephemeral_1h_input_tokens":27892,"ephemeral_5m_input_tokens":0},"inference_geo":"not_available","iterations":{"input_tokens":3,"output_tokens":2360,"cache_read_input_tokens":39148,"cache_creation_input_tokens":5857,"cache_creation":{"ephemeral_5m_input_tokens":0,"ephemeral_1h_input_tokens":5857},"type":"message"}],"speed":"standard"},"modelUsage":{"claude-haiku-4-5-20251001":{"inputTokens":10508,"outputTokens":15,"cacheReadInputTokens":0,"cacheCreationInputTokens":0,"webSearchRequests":0,"costUSD":0.010583,"contextWindow":200000,"maxOutputTokens":32000,"sonnet-4-6":{"inputTokens":48,"outputTokens":40418,"cacheReadInputTokens":2218994,"cacheCreationInputTokens":129443,"webSearchRequests":0,"costUSD":1.8202804500000005,"contextWindow":200000},"terminal_reason":"completed","fast_mode_state":"off","uuid":"7593cbd6-8ea3-4259-bda9-2832cc931f9c"}}